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Studierichting: Informatica

Oefeningenreeks 4A nr. 66.1

$$\begin{aligned} & \int \sqrt{x^2 - 1} dx \stackrel{\sqrt{x^2 - 1} = x + t}{=} \int \left(-\frac{t^2 + 1}{2t} + t \right) \cdot \left(-\frac{t^2 - 1}{2t^2} \right) dt \\ &= \int -\frac{(t^2 - 1)^2}{4t^3} dt = -\frac{1}{4} \int \frac{t^4 - 2t^2 + 1}{t^3} dt \\ &= -\frac{1}{4} \left(\int t dt - \int \frac{2}{t} dt + \int \frac{1}{t^3} dt \right) \\ &= -\frac{1}{4} \left(\frac{t^2}{2} - 2 \cdot \ln |t| - \frac{1}{2t^2} \right) + c \\ &= -\frac{t^2}{8} + \frac{\ln |t|}{2} + \frac{1}{8t^2} + c \\ &\stackrel{t = \sqrt{x^2 - 1} - x}{=} -\frac{(\sqrt{x^2 - 1} - x)^2}{8} + \frac{\ln |\sqrt{x^2 - 1} - x|}{2} + \frac{1}{8(\sqrt{x^2 - 1} - x)^2} + c \end{aligned}$$

References